

**MATH 2101 LINEAR ALGEBRA I, FALL 2025 – OUTLINE OF
SOLUTION FOR ASSIGNMENT 4**

Due date: 10 October (Friday), 2025 10:00pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
 - All solutions have to be turned in HKU moodle in the format of *PDF file*.
 - Please include your *Name, UID, Faculty, Major (if declared)* in your solution.
1. For two vector subspaces W_1, W_2 of a vector space V , define

$$W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\}.$$

Show that $W_1 + W_2$ is a vector subspace of V .

Proof. One shall check the axioms for vector spaces one by one (the labeling below follows the lecture note). I will demonstrate some verifications and leave the easy parts to you.

For any $w_1 + w_2, v_1 + v_2 \in W_1 + W_2$, where $w_i, v_i \in W_i$ for $i \in \{1, 2\}$, and $r \in \mathbb{R}$,

- Addition:

$$(w_1 + w_2) + (v_1 + v_2) = (w_1 + v_1) + (w_2 + v_2) \in W_1 + W_2.$$

- Scalar multiplication:

$$r.(w_1 + w_2) = r.w_1 + r.w_2 \in W_1 + W_2.$$

- (1)-(3) Commutativity, associativity of addition, and the existence of the zero element are inherited from the ambient vector space V .

- (4) For any $w_1 \in W_1$ and $w_2 \in W_2$, the inverse of $w_1 + w_2 \in W_1 + W_2$ is $-w_1 - w_2$:

$$(w_1 + w_2) + (-w_1 - w_2) = w_1 - w_1 + w_2 - w_2 = 0 \in W_1 + W_2.$$

- (5) For any $w_1 + w_2 \in W_1 + W_2$, $1 \in V$, $1.(w_1 + w_2) = 1.w_1 + 1.w_2 = w_1 + w_2$.

- (6) For any $a, b \in \mathbb{R}$, $w_1 \in W_1$, $w_2 \in W_2$,

$$a(b(w_1 + w_2)) = a(bw_1 + bw_2) = a(bw_1) + a(bw_2) = (ab)w_1 + (ab)w_2 = (ab)(w_1 + w_2).$$

- (7)-(8) These can be checked similarly to axiom (6).

□

2. Let S be a subset of a vector space V . Show that $\text{span}(S) = S$ if and only if S is a vector subspace of V .

Proof. If $\text{span}(S) = S$, then S is a vector subspace of V because it is a spanning set. Conversely, if S is a vector subspace of V , then by definition, one has an inclusion $S \subset \text{span}(S)$ of vector spaces. For any $u \in \text{span}(S)$, we can write $u = a_1 s_1 + \cdots + a_r s_r$ for $a_i \in \mathbb{R}$ and $s_i \in S$ for $i \in \{1, \dots, r\}$. Since S is a vector space, addition and scalar multiplication preserve the elements in S . It follows that $u \in S$, which shows the reversed inclusion. $\square$

3. Let $\text{Mat}_{5 \times 5}$ be the vector space of 5×5 matrices. Determine if the following sets form a vector subspace of $\text{Mat}_{5 \times 5}$.

- (a) Let B be a 6×5 -matrix. Let $S_1 = \{A \in \text{Mat}_{5 \times 5} : BA = 0\}$.
- (b) Let $S_2 = \{A \in \text{Mat}_{5 \times 5} : \det(A) = 0\}$.
- (c) Let $S_3 = \{A \in \text{Mat}_{5 \times 5} : A^T = A\}$.
- (d) Let B be a 5×5 -matrix. Let $S_4 = \{A \in \text{Mat}_{5 \times 5} : BA = I_5\}$.

Explain your answers.

Sol.

- (a) It is a vector subspace. For any $A_1, A_2 \in S_1$ and $r \in \mathbb{R}$, we have $B(A_1 + A_2) = BA_1 + BA_2 = 0$ and $B(rA_1) = rBA_1 = 0$. In other words, $A_1 + A_2 \in S_1$ and $rA_1 \in S_1$.
- (b) It is not a vector subspace. Consider two diagonal matrices $A_1 = \text{diag}(1, 0, 0, 0, 0)$ and $A_2 = \text{diag}(0, 1, 1, 1, 1)$. Note that A_1, A_2 has zero determinant but $\det(A_1 + A_2) = \det(I_5) = 1$, i.e., $A_1 + A_2 \notin S_2$. Hence, addition is not closed in S_2 .
- (c) It is a vector subspace. For any $A_1, A_2 \in S_3$ and $r \in \mathbb{R}$, we have $(A_1 + A_2)^T = A_1^T + A_2^T = A_1 + A_2$ and $(rA_1)^T = rA_1^T = rA_1$. In other words, $A_1 + A_2 \in S_3$ and $rA_1 \in S_3$.
- (d) It is not a vector subspace. Consider any $A \in S_4$, we have $BA = I_5$. However, $2A$ satisfies $B(2A) = 2BA = 2I_5 \neq I_5$. Hence, scalar multiplication is not closed in S_4 .

4. Determine whether $\begin{pmatrix} 1 \\ 0 \\ 2 \\ 3 \end{pmatrix}$ is in

$$\text{span}\left(\left\{\begin{pmatrix} 1 \\ -2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}\right\}\right).$$

Explain your answer.

Sol. I claim the vector is not in the spanning set. Assume to the contrary that it is, then the following linear system

$$\begin{pmatrix} 1 & 1 \\ -2 & 1 \\ 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \\ 3 \end{pmatrix}$$

has a solution. To check whether the system has a solution or not, we consider the

associated augmented matrix $\begin{pmatrix} 1 & 1 & | & 1 \\ -2 & 1 & | & 0 \\ 1 & -1 & | & 2 \\ 1 & -1 & | & 3 \end{pmatrix}$. It has RREF $\begin{pmatrix} 1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ 0 & 0 & | & 1 \\ 0 & 0 & | & 0 \end{pmatrix}$,

which is inconsistent. So there is no solution, a contradiction.